Universality of boundary electrostatics for forced defects in dimer coverings: entropic screening without bipartiteness and corner charges on the honeycomb lattice

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In a companion paper we showed that forced defects (monomers) in square-lattice dimer coverings behave as charges of a two-dimensional Coulomb gas: every corner carries a fractional charge fixed by the jump of the boundary height, and a free boundary produces images of the same sign. Here we ask which parts of this picture are universal. We test it in three directions, using exact Pfaffian and Kasteleyn computations on regions of up to 1.5×10^5 sites. (i) Stiffness. Because defects and corner charges are both vortices of the height field, the whole density scales as $\rho_g \propto \rho_{1/2}^{2g}$, so every exponent is multiplied by the same factor $2g$; the corner exponent becomes $\gamma(g) = 2g[(\pi/\theta)(\frac{1}{2} + eq_c) - \frac{1}{4}]$. Worm Monte Carlo of interacting dimers confirms that the edge and bulk pair exponents keep the ratio 2.01 ± 0.05 while g varies twentyfold. (ii) Bipartiteness. Adding diagonal bonds of weight t interpolates between the square and triangular lattices, where the bulk monomer correlation is known exactly. At the boundary, trapping stops at $N \sim 1/t$ and the saturated corner ratio depends on the corner angle; an apparent exponent $\nu \approx 1.26$ in the corner ratio is traced to a large $N \ln N$ correction, the true value being $\nu = 1$. At the critical point $t = 1/2$, which the existing theorems exclude, the pair correlation takes the closed form $[1 - (\sqrt{3} - 2)^{|\Delta r| + |\Delta c|}]/6$, which we derive from the fact that the block-Toeplitz symbol becomes rational there. (iii) Lattice. On the honeycomb lattice, armchair boundaries give flat liquids and zig-zag boundaries freeze. The monomer winding is $w = 3$, and at 60° armchair corners the rule $q_c = -J_c/3$ holds exactly and predicts the corner exponent $\gamma = 1/4$. At 120° corners the naive rule fails. We show that the lattice boundary layer shifts the continuum boundary height by $\pm 1/4$ on the two classes of armchair edges (a symmetry fixes the opposite signs, and an exact infinite-strip calculation gives the size to 5×10^{-4}), so that $q_c = -(J_c + \sigma/2)/3$ with $\sigma = \pm 1$. Fixed on hexagons, this rule predicts all eight corner exponents of armchair parallelograms blind, to within 0.05. Same-sign images persist on armchair edges.

I. INTRODUCTION

Close-packed dimer models are the simplest lattice systems whose long-distance physics is a free field [2–8]. The height field h that encodes the close-packing constraint fluctuates like a Gaussian free field, and monomers are its vortices. When the region cannot be tiled completely, monomers are forced to appear [Fig. 1(a)], and the question of where they sit becomes a problem of two-dimensional electrostatics [Fig. 1(b)].

In the companion paper (Paper I) [1] we showed, for the square lattice at the free-fermion point, that this electrostatics has three ingredients. First, every corner of the region carries a fractional charge $q_c = -J_c/4$, where J_c is the jump of the boundary height at the corner. Second, a free boundary acts as a mirror that produces images of the same sign. Third, frozen boundaries add a boundary chemical potential. With these rules a conformally covariant density formula,

$$\rho(z) \propto |f'(z)|^{1/4} \, (\mathrm{Im}\, f)^{1/4} \prod_c |f(z) - f_c|^{eq_c}, \qquad (1)$$

where f maps the region onto the upper half plane and $e = \pm 1$ is the defect charge, reproduced exact distributions without adjustable parameters.

Much of the bulk side of this picture is established. Ciucu and co-workers proved that fixed gaps interact by two-dimensional electrostatics, first for lozenge tilings [11], then on the square lattice [13, 14] and on all doubly periodic bipartite lattices in the liquid phase [15]; near boundaries the image of a gap keeps its sign at constrained boundaries and reverses it at free ones [12, 16, 17]. The boundary-height jumps at corners were identified by Allegra as boundary-condition-changing operators of dimension $J^2/32$ [18]. For non-bipartite lattices, the monomer correlation of the square-to-triangular interpolation is known exactly [9, 19, 20]. Bulk universality is therefore not in question here; what we ask concerns corners, boundaries, and defects that are free to move.

The present paper asks which parts of this structure are universal. We test it in three directions. The stiffness g can be changed by interactions; bipartiteness can be broken by diagonal bonds; and the lattice can be changed from square to honeycomb at fixed stiffness. The results are organized around a single question for each direction: what survives, and what replaces what does not.

Our main results are as follows. The stiffness enters only as an overall factor, so all exponents scale together (Sec. II). Without bipartiteness the entropic Coulomb force is screened beyond an exactly computable length, and forced defects deconfine (Sec. III). On the honeycomb lattice the corner-charge idea survives, but at 120° armchair corners it acquires a new ingredient, a boundary-

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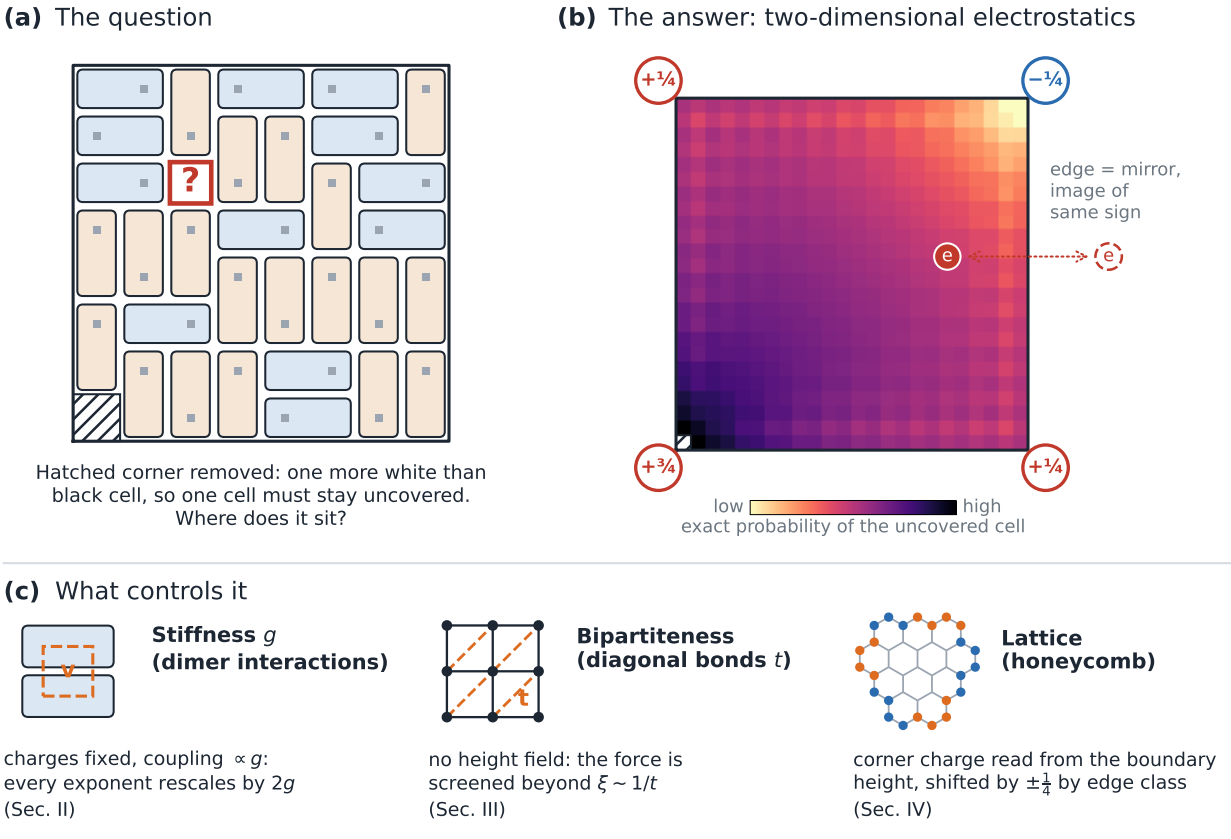


FIG. 1. The problem and its electrostatic answer. (a) An 8×8 board with one corner removed has one more white than black cell, so every dimer covering leaves one white cell uncovered (a sample covering; grey squares mark black cells). Where is that cell? (b) Exact probability of the uncovered cell on a 24×24 board with the same corner removed. It is drawn to the removed corner and pushed away from the opposite one, as a charge $e = -1$ would be in the field of fixed charges at the corners ($q_c = +\frac{3}{4}$ at the removed corner, $\pm\frac{1}{4}$ at the others), with each edge acting as a mirror that produces an image of the same sign. (c) The three controls studied here: the stiffness of the height field, changed by interactions between dimers; bipartiteness, broken by diagonal bonds; and the lattice, changed to the honeycomb.

layer shift of the continuum boundary height that depends on the class of the edge (Sec. IV). This rule passes a blind test on regions with both 60° and 120° corners. Section V collects the verdicts in one table.

II. STIFFNESS DEPENDENCE OF THE COULOMB-GAS FORMULA

We use the conventions of Paper I: the normalized height $h = h_T/2$, where h_T is Thurston's height; a monomer winds h_T by ± 4 and creates $(2/\pi)\ln|z - z_0|$ in the dual field; a boundary height jump J at c creates $(J/\pi)\ln|z - c|$. At the free-fermion point the stiffness is $g = 1/2$.

The key observation is that both the defect and the corner charges are vortex-type (magnetic) charges, and that the corner charges are fixed by topology: $q_c = -J_c/4$ does not depend on g. For a fixed configuration of such charges the Gaussian weight is $\exp(-g\,E)$, where E depends only on geometry (positions, the conformal map, and the cutoff self-energy that produces the $|f'|$ factor). Hence, within the Gaussian theory,

$$\rho_g \propto \rho_{1/2}^{2g}, \qquad \rho_g \propto |f'|^{g/2}(\mathrm{Im}\, f)^{g/2} \prod_c |f - f_c|^{2g\, e q_c}. \tag{2}$$

The monomer scaling dimension is $x_m = g/2$, and near a corner of interior angle θ, where $f \sim z^{\pi/\theta}$,

$$\gamma(g) = 2g\left[\frac{\pi}{\theta}\left(\frac{1}{2} + e q_c\right) - \frac{1}{4}\right]. \tag{3}$$

At $g = 1/2$ this reduces to the corner exponent of Paper I, and it reproduces all of its exponents: $-3/4$ at a removed corner, $N^{-1/4}$ for the corner-to-centre ratio of odd boards, $d^{-1/2}$ for bulk pairs and d^{-1} for pairs along a free edge (in general $-3g/2$, $N^{-g/2}$, d^{-g} and d^{-2g}).

The reproduction at $g = 1/2$ is automatic. The content of Eq. (2) is the prediction that *all* exponents are

rescaled by the same factor, so that their ratios are independent of g. This is a parameter-free test for interacting dimers. Our g coincides with that of Alet *et al.* [10], whose Coulomb-gas exponents are $\alpha(e,m) = e^2/g + g\,m^2$, with monomer correlations decaying as r^{-g} and $g = 1/2$ at infinite temperature; aligning interactions raise g up to $g = 4$ at the columnar transition. The assumptions behind Eq. (2) are that the dual field keeps Neumann conditions at free edges for all g and that no height-locking operator is relevant (which fails near the columnar transition).

Monte Carlo test. We simulated the interacting dimer model of Ref. [10], with weight $e^{-\beta v N_\parallel}$ ($N_\parallel$ is the number of plaquettes with two parallel dimers), on a cylinder of circumference 64 and height 128. A worm whose tail is fixed and whose head moves samples $Z(\text{tail},\text{head})$ directly; the algorithm reproduces exhaustive enumeration on 4×4 and 6×4 cylinders to 3×10^{-4} for $\beta v = 0, \pm1$. We placed the tail in the bulk or on the open edge and measured the pair weight along the same row against the chord distance (Fig. 2). Equation (2) predicts d^{-g} in the bulk and d^{-2g} along the edge, i.e. a ratio of exactly 2 for every g. For $\beta v = +1, +0.5, 0, -0.4, -0.7, -1$ the bulk exponent is $g = 0.092$, 0.243, 0.506, 0.839, 1.224, 1.880 (exact 1/2 at $\beta v = 0$), and the ratio is 2.11, 1.99, 1.95, 1.98, 2.01, 2.00, i.e. 2.01 ± 0.05 over a twentyfold change of g. This also confirms that the dual field keeps its Neumann condition, i.e. same-sign images, at all g.

III. BREAKING BIPARTITENESS

A. Model and exact method

We add a bond of weight t across one diagonal of every square plaquette [Fig. 3(a)]. At $t = 0$ this is the square lattice; at $t = 1$ it is the triangular lattice. The graph is planar but not bipartite for $t > 0$, so we use Pfaffians. Requiring an odd number of clockwise edges on every triangular face gives a Kasteleyn orientation: horizontal edges point east, vertical edges alternate up and down from column to column, and each diagonal follows the vertical edge of the column to its right. A monomer in the interior is handled by reversing the signs of the edges crossed by a string running downward to the boundary. The orientation was checked against exact transfer-matrix and brute-force counts: single monomers at every site of $N\times N$ boards with $N = 5, 7, 9$ and all monomer pairs on 4×4 and 6×6 boards agree exactly. Large systems use sparse LU factorizations of the string-modified matrix, which avoid the cancellations of Schur-complement updates (Appendix A).

B. Boundary trapping stops at $N \sim 1/t$

On odd $N\times N$ boards one monomer is forced. Figure 4 shows the ratio R of its probability at corner A to that at the centre. At $t = 0$ we recover $R \propto N^{-1/4}$; the two branches $N \equiv 1, 3 \pmod 4$ converge toward each other ($RN^{1/4} = 1.831$ and 1.900 at $N = 69, 71$). For $t > 0$, R saturates once N exceeds roughly $1/t$: the entropic force that trapped the defect near the corner disappears. The saturated values remember the corner angle (Table I): the 60° corner B attracts more than the 120° corner A, so non-universal boundary-layer information survives screening.

C. Deconfinement and the decay of pair correlations

Figure 5(a) shows the bulk pair function $C(r) = Z(\text{two monomers})/Z$. At $t = 0$ it decays as a power; for $t > 0$ it tends to a non-zero constant, i.e. the monomers deconfine [9]. The limit is known in closed form [19],

$$C_\infty = \frac{1}{2}\sqrt{\frac{t}{2t(2+t^2)+(1+2t^2)\sqrt{2+t^2}}}, \qquad (4)$$

which gives 1/6 exactly at $t = 1/2$; our values at $r = 80$ agree with it to 10^{-13} (3×10^{-10} at $t = 0.1$), an independent check of the numerics.

To determine how C_∞ is approached we computed $C(r)$ up to $r = 80$ on 256×256 boards (384×384 for $t = 0.1$), with values independent of the board size to 10^{-12}. For $t \le 0.3$,

$$C(r) - C_\infty \simeq (-1)^{r+1} A(t)\, r^{-1} e^{-2\kappa r}, \qquad (5)$$

with $2\kappa = 2\ln\eta_2(t)$, the rate proved by Basor and Bleher for $0 < t < 1/2$ [20]; here $\eta_2^{-2} = 2 - \mu - 2\sqrt{1-t^2-\mu}$, $\mu = \sqrt{1-4t^2}$, and $\ln\eta_2 = t + O(t^3)$. Our data reproduce the r^{-1} prefactor and the alternating component [Fig. 5(b)]: with the exact C_∞, the residual rate δ_0 in a fit $\alpha/r + \delta_0$ vanishes within 0.003 for $2\ln\eta_2$, while $2\,\mathrm{artanh}\,t$, which agrees with it at small t, leaves $\delta_0 = 0.006$, 0.017 and 0.059 at $t = 0.35$, 0.4 and 0.45. For $t > 1/2$ the two rates $\eta_{1,2}$ become complex and the decay oscillates incommensurately, as we also observe.

D. Exact screening length and the apparent exponent

With a two-column unit cell the Kasteleyn symbol is

$$P(\theta,\varphi) = 4\sin^2\varphi + \left|1 - e^{-i\theta} - t\,e^{i\varphi} - t\,e^{-i\theta-i\varphi}\right|^2, \quad (6)$$

TABLE I. Saturated corner-to-centre ratios R_∞ ($N = 61$).

t	corner A (120°)	corner B (60°)
0.2	1.087	1.172
0.5	1.350	1.732
1.0	1.526	2.587

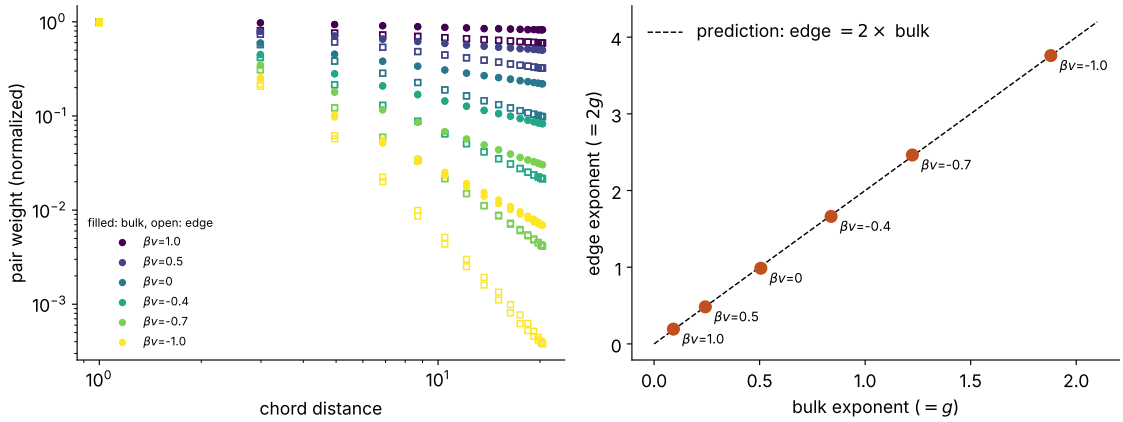


FIG. 2. Worm Monte Carlo of interacting dimers. (a) Pair weight against chord distance for the tail in the bulk (filled) and on the open edge (open), for several βv. (b) Edge exponent against bulk exponent; the dashed line is the prediction $2g$ against g from Eq. (2).

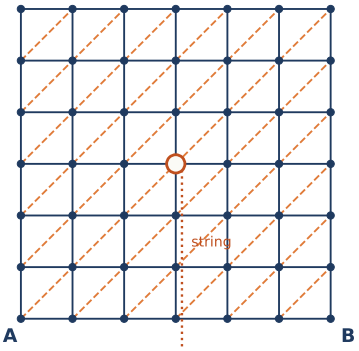


FIG. 3. (a) Square lattice with diagonal bonds of weight t (dashed). A monomer (open circle) is attached to the boundary by a string (dotted) along which edge signs are reversed. The corners A (touched by a diagonal) and B become 120° and 60° corners of the triangular lattice.

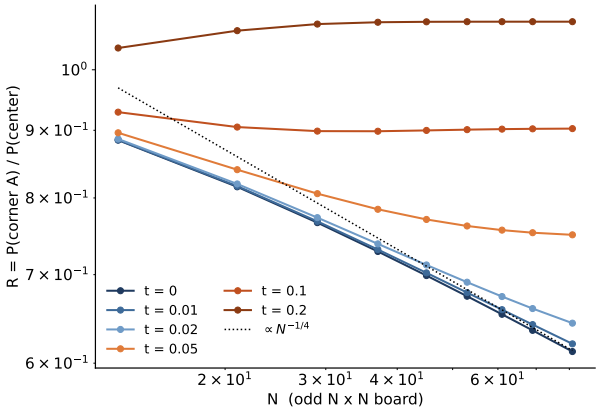


FIG. 4. Corner-to-centre ratio R of a forced monomer on odd $N \times N$ boards ($N \equiv 1 \pmod 4$). At $t = 0$, $R \propto N^{-1/4}$ (dotted); for $t > 0$, R saturates at $N \sim 1/t$.

which has no zero on the unit torus for $t > 0$: the perturbation gives a mass to the two Majorana fermions of the square lattice [9]. The $\varphi = 0$ branch of its zeros gives a decay rate $\operatorname{artanh} t$ per site [Fig. 6(a)]; the monomer correlation length is $1/(2 \ln \eta_2) \simeq 1/(2t)$ [20]. Either way the screening length diverges as $1/t$, i.e. $\nu = 1$.

A naive scaling collapse of the corner ratio, $R(N,t)/R(N,0) = G(N/\xi)$, instead gives $\nu \approx 1.26$ (the same for both corners when $N \geq 53$). The resolution is an exact symmetry. Diagonal bonds join sites of the same colour, so once the monomer position is fixed the parity of the number of diagonal dimers is fixed, and $Z(-t) = \pm Z(t)$. Hence $\ln R(N,t) - \ln R(N,0) = a_2(N) t^2 + O(t^4)$. We find

$$a_2(N) = cN^2 + eN \ln N + f, \qquad c \approx 0.015, \qquad (7)$$

with the same c for both corners. The leading term $(Nt)^2$ is the one expected for $\nu = 1$; the subleading coefficient is some thirty times larger than c, which produces the apparent exponent. A fit on $N \leq 81$ predicts $a_2(101)$ to 0.1% and 0.01% for the two corners, whereas a pure power law misses by 1.3–3.3% [Fig. 6(b)].

E. A closed form at $t = 1/2$

At $t = 1/2$ the pair function is, to 10^{-10} accuracy,

$$C(\Delta r, \Delta c) = \frac{1}{6}\left[1 - (\sqrt{3} - 2)^{|\Delta r| + |\Delta c|}\right] \qquad (\Delta r\, \Delta c \geq 0), \qquad (8)$$

checked along $(1,0)$, $(0,1)$, $(1,1)$ and $(1,2)$ (Fig. 7); the anti-diagonal direction $(1,-1)$ is different. The point $t = 1/2$ is the critical value at which the two rates η_1, η_2 of Basor and Bleher merge, and it is excluded from

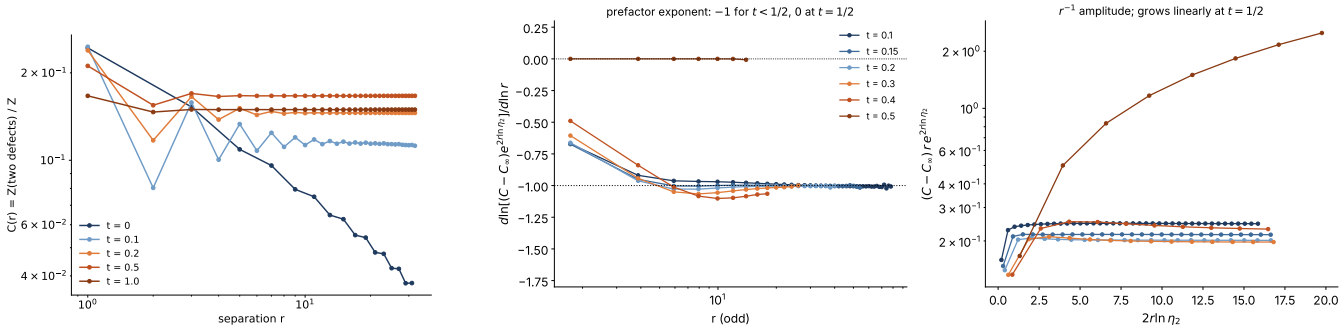


FIG. 5. (a) Bulk pair function $C(r)$ on 64×64 boards; at $t = 0$ even separations vanish by the colour constraint. (b) Local prefactor exponent of $(C - C_\infty)e^{2r \ln \eta_2}$ (odd r, exact C_∞); it tends to -1 for $t < 1/2$ and to 0 at the critical point $t = 1/2$. (c) $(C - C_\infty)\, r\, e^{2r \ln \eta_2}$: flat for $t < 1/2$, growing linearly at $t = 1/2$ where the r^{-1} prefactor is absent.

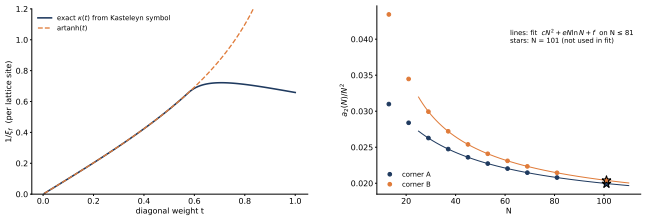


FIG. 6. (a) Exact fermion decay rate from the Kasteleyn symbol and $\operatorname{artanh} t$. (b) $a_2(N)/N^2$ for both corners; lines are fits $cN^2 + eN \ln N + f$ on $N \le 81$, stars are $N = 101$ (not used in the fit).

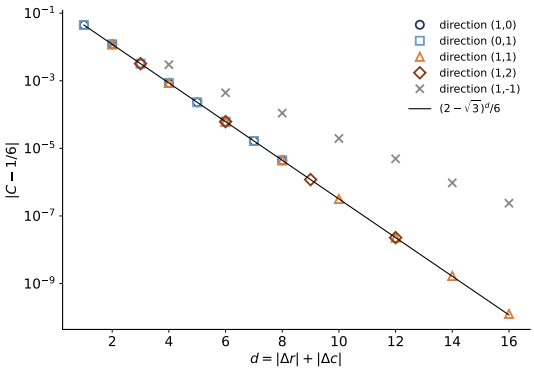


FIG. 7. $|C - 1/6|$ at $t = 1/2$ against the Manhattan distance for several directions; the line is $(2 - \sqrt{3})^d/6$.

their theorems [20]. Their rate gives $\eta_2^{-2} = 2 - \sqrt{3}$ there, i.e. $2 \ln \eta_2 = \ln(2 + \sqrt{3})$, which is exactly the rate of our closed form, and the $C_\infty = 1/6$ of Basor and Ehrhardt fixes its constant. What the closed form adds is that at the critical point the correction is a pure exponential at all distances, with no power-law prefactor. This follows from the structure of the symbol. In the representation $C = \frac{1}{2}\sqrt{\det T_n(\phi)}$ of Ref. [20], ϕ contains $[t^2 + \sin^2 x + \sin^4 x]^{-1/2}$; with $u = \sin^2 x$ the radicand $u^2 + u + t^2$ has discriminant $1 - 4t^2$. For $t < 1/2$ its two roots produce square-root branch points, whose Fourier coefficients decay as $k^{-1/2}\eta_2^{-k}$ and give the $r^{-1}e^{-2r \ln \eta_2}$ correction. At $t = 1/2$ the roots merge, $u^2 + u + \frac{1}{4} = (u + \frac{1}{2})^2$, the square root disappears and the symbol becomes rational (likewise $\sqrt{2 + t^2} = 3/2$ in C_∞). For a rational symbol $\det T_n$ is a finite sum of exponentials in n. Evaluating the block determinant with 40-digit arithmetic gives $C = [1 - (\sqrt{3} - 2)^{n+1}]/6$ for $n = 1, \ldots, 30$ to 10^{-41}, and $4C^2$ satisfies a linear recurrence of order exactly three. Near $t = 1/2$ the amplitude of the r^{-1} term carries $1/\sqrt{\eta_1^2 - \eta_2^2}$ with $\ln(\eta_1/\eta_2) \propto (1/2 - t)^{1/2}$, so the r^{-1} form holds only beyond a crossover length $\propto (1/2 - t)^{-1/2}$ that diverges at the critical point.

IV. THE HONEYCOMB LATTICE

A. Heights, flat regions and forced defects

Honeycomb dimers are lozenge tilings, for which Ciucu established the bulk Coulomb law of holes [11]; the bulk law holds on every doubly periodic bipartite lattice [15], so the questions below concern only boundaries and corners. In Thurston units (an empty edge contributes $+1$ and a dimer -2 around a vertex of degree three) a monomer winds the height by ± 3, so the corner rule generalizes to $q_c = -J_c/3$. The monomer is still a unit magnetic charge, and $g = 1/2$ in the normalization of Sec. II.

Paper I relied on regions whose limit shape is flat [21]. In brick-wall form the honeycomb is the square lattice with one of the four dimer types missing, so its slope polygon is a triangle, and the zero tilt forced by an axis-parallel rectangle lies on an edge of that triangle. Flat liquids therefore need boundaries whose tilt is interior. Armchair boundaries provide them. For D_6-symmetric armchair flakes the local deviation of the bond probabilities from 1/3 decreases with size (core mean

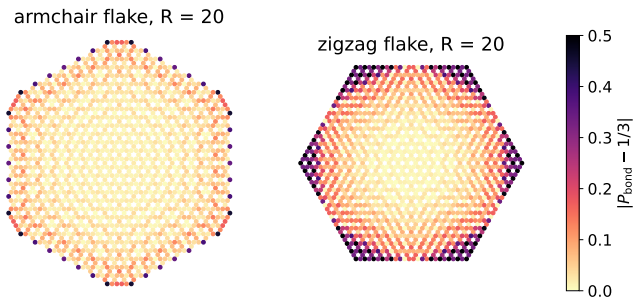


FIG. 8. Local deviation $|P-1/3|$ of the exact bond probabilities. Armchair flakes are flat liquids; zig-zag flakes (lozenge hexagons) freeze near the corners.

$|P-1/3| = 0.035,\ 0.017,\ 0.012$ for $R = 12, 20, 28$), with no nearly frozen bond; zig-zag flakes, which are lozenge hexagons, freeze near the corners (Fig. 8). (Orientation-averaged annular means hide this difference, because the D_6 symmetry makes each annulus average 1/3.)

Armchair flakes centred on an A site have $n_A = n_B + 1$, so one A defect ($e = +1$) is forced without removing any site; the same holds for all armchair triangles centred on an A site, and for suitable parallelograms [Fig. 9]. Interior defects are again attached to the boundary by strings; the method reproduces brute-force counts at every defect site of small flakes.

B. Boundary heights and the naive corner rule

Walking around the boundary on the dual triangular lattice, the height changes by $+1$ (-1) when the interior vertex next to the step is A (B). Along an armchair edge the steps alternate, so the midpoint height is exactly constant; jumps occur only at corners (Fig. 10). Because the defect winds the height by 3, the jumps are defined modulo 3 (the analogue of the equivalence $-3 \equiv +1$ in Paper I). Hexagons fall into two families by corner termination: type I with $J = 0, +1$ on alternating corners and type II with $J = -1, +2$. In both, $\sum_c J_c = 3$.

Type I hexagons turn out to be Temperleyan [22]: all boundary A sites have exactly the same defect probability, and the profiles entering from a $J = 0$ and a $J = +1$ corner coincide to three digits although the two corners are geometrically different. The A sublattice of these flakes is sixfold symmetric, and the defect density inherits that symmetry [Fig. 11]. Type II hexagons are not Temperleyan.

The corner exponents, from corner-to-centre ratios $\propto R^{-\gamma}$ (Fig. 12), are $\gamma \approx 0.28$ at both corners of type I, and $\gamma \approx 0.76$ and -0.25 at the two corners of type II. The naive rule $q_c = -J/3$ with Eq. (3) predicts 1/2 and 0 for type I, and 1 and $-1/2$ for type II. It fails at every 120° corner.

C. 60° corners: a rule-independent test

Armchair triangles centred on an A site carry one forced defect for every size ($R = 6$–59). Their three corners are related by the C_3 symmetry, so neutrality alone fixes $q_c = -1/3$, whatever rule is used for J. The naive jump is $J = +1$ at all three corners, so $q_c = -J/3$ is exact here. Equation (3) with $\pi/\theta = 3$ predicts $\gamma = 1/4$ (5/4 without corner charge). The measured value is $\gamma = 0.22$ from all sizes, and the local exponents of same-phase subsequences converge to it (0.19, 0.22, 0.23, 0.25 for $R = 8$, 16, 24, 32, 40). This geometry is of the type solved by Ciucu and Fischer [17], whose "zig-zag" sides on the triangular lattice are our armchair edges; their result corresponds to $q_c = 0$ and ours extends it to charged corners.

D. The edge-class shift

Armchair edges come in two classes by the direction of their outward normal: $0°, 120°, 240°$ (class +) and $60°, 180°, 300°$ (class −). A 60° rotation exchanges the classes and the sublattices, which reverses the sign of the height. The lattice boundary layer therefore shifts the continuum boundary value by $+\delta$ on class + edges and by $-\delta$ on class − edges. A 120° corner joins edges of different classes, so its effective jump is $J \pm 2\delta$; a 60° corner joins edges of the same class and is unaffected. With $2\delta = 1/2$,

$$q_c = -\frac{J_c + \sigma_c/2}{3}, \tag{9}$$

where $\sigma_c = +1$ (-1) if, going counterclockwise, the corner leads from a class + (class −) edge to a class − (class +) edge, and $\sigma_c = 0$ at 60° corners. In Allegra's language [18] the edge-class shift changes the boundary-condition-changing operator at a 120° corner from the one set by the lattice jump J to the one set by $J \pm 1/2$. For type I hexagons this gives $q_c = -1/6$ at all six corners and $\gamma = 1/4$, i.e. the Temperleyan density $\rho \propto \mathcal{R}^{1/4}$ of the conformal radius; for type II it gives $+1/6$ and $-1/2$, i.e. $\gamma = 3/4$ and $-1/4$. In both families $\sum_c q_c = -1$.

Blind test. Armchair parallelograms have two 60° and two 120° corners. With Eq. (9) fixed on hexagons, we predicted their exponents before computing them (Table II). All eight corners agree within 0.05, while the naive rule misses every 120° corner by 0.25–0.3. Figure 13 collects all thirteen corners studied.

Independent determinations of the shift. Two further measurements support Eq. (9) without fitting J. First, in balanced flakes (no defect) the exact mean height $\langle h \rangle$ follows from $\langle \Delta h \rangle = 1 - 3P$ on every bond. The continuum boundary value of an edge is the value the interior mean field approaches, so we fit the interior to $c + a\,u(z)$, with u the harmonic function on the hexagon with boundary values ± 1 on alternating edges. The resulting $J_{\rm eff}$ lies near the predicted half-integers, far from the naive integers [Fig. 14(a)]; the $J = 0$ family approaches $+1/2$ as

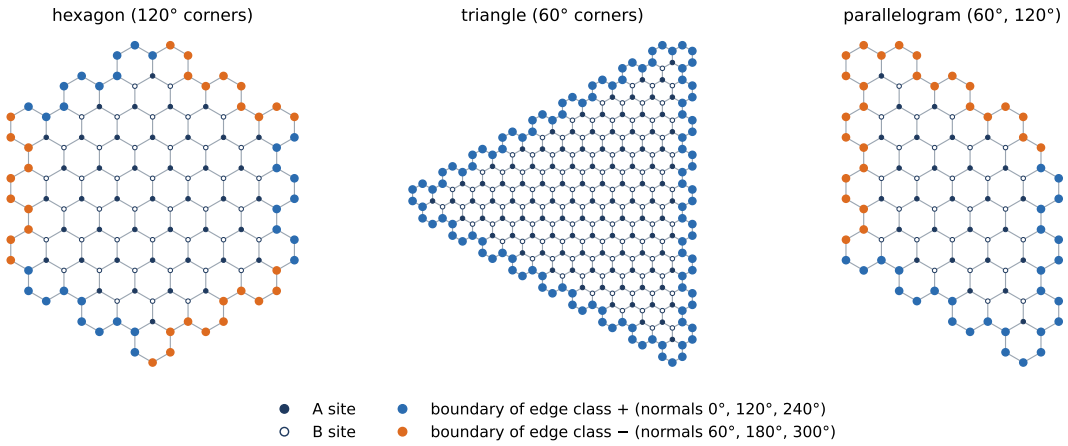


FIG. 9. Armchair regions used in Sec. IV, with boundary sites coloured by the class of their edge. Hexagons have only 120° corners, each joining edges of different classes; triangles have only 60° corners, joining edges of the same class; parallelograms have both.

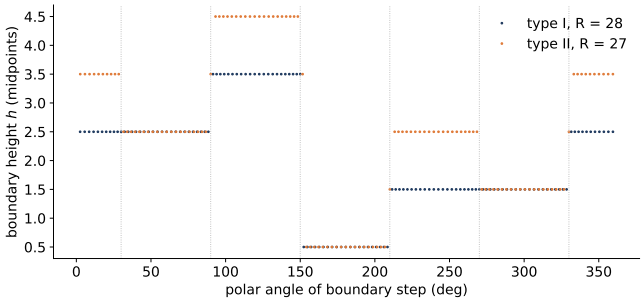


FIG. 10. Boundary height (step midpoints) around type I and type II hexagons with one forced defect. The height is constant along each edge and jumps at the corners (dotted lines); the jump of −3 is the branch cut of the defect.

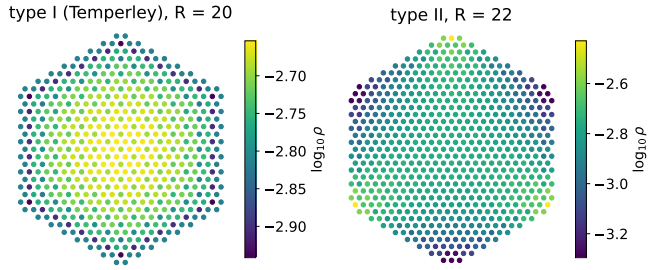


FIG. 11. Exact defect density on A sites of a type I (left) and a type II (right) hexagon.

R grows. Second, a 180° rotation about a hexagon centre exchanges the sublattices and the two edge classes, so $\delta_- = -\delta_+$ exactly. The size follows from an infinite armchair strip, solved exactly by Fourier transformation along the edge: each wave number gives a finite Kasteleyn matrix, and the bond probabilities are integrated with convergence checked by doubling the grid. Extrapolating the linear interior profile of $\langle h \rangle$ to the two edges gives $\delta_+ - \delta_- = -0.402$, -0.484, -0.489, -0.505, -0.502 and -0.4995 for half-widths $R = 12$, 20, 40, 60, 120 and 240 [Fig. 14(b) shows the earlier finite-strip estimates]; the deviation from $-1/2$ falls roughly as $1/R$, to 5×10^{-4}. That the shift is exactly $1/2$ follows from the Temperley property of type I hexagons: equal boundary probabilities require $eq_c = -1/6$ at all six corners.

TABLE II. Blind test on armchair parallelograms: predicted and measured corner exponents (measured from the largest five sizes). J is the exact naive jump with the branch cut fixed at the 120°-direction corner.

family	corner	J	Eq. (9)	naive	measured
F1	30° (120° angle)	−1	+0.75	+1.00	+0.71
F1	120° (60° angle)	+1	+0.25	+0.25	+0.24
F1	210° (120° angle)	+2	−0.25	−0.50	−0.27
F1	300° (60° angle)	+1	+0.25	+0.25	+0.24
F2	30°	0	+0.25	+0.50	+0.28
F2	120°	+1	+0.25	+0.25	+0.28
F2	210°	+1	+0.25	0.00	+0.28
F2	300°	+1	+0.25	+0.25	+0.28

E. Same-sign images on armchair edges

At the midpoint of an armchair edge, a same-sign image predicts a defect probability $\propto R^{-1/4}$ relative to the centre and a profile $\propto d^{+1/4}$ into the bulk; an opposite-sign image would give $R^{+1/4}$ and $d^{-1/4}$. We find $R^{-0.29}$ (type I) and $R^{-0.24}$ (type II), and inward slopes +0.20 and +0.21 over $2 < d < R/4$, where the $\mathrm{Im}\, f$ factor starts to saturate (Fig. 15). Armchair edges are thus constrained boundaries in Ciucu's sense, for which im-

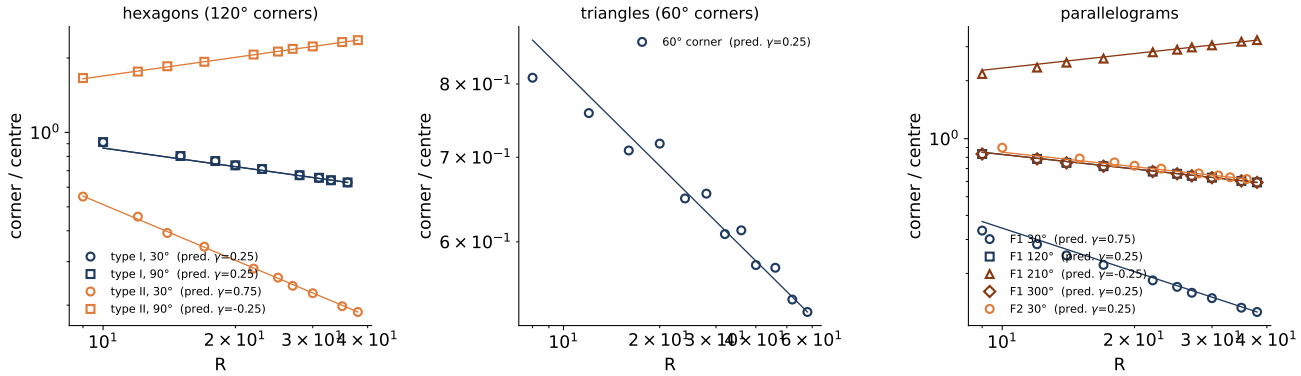


FIG. 12. Corner-to-centre ratios against size for hexagons, triangles and parallelograms. Lines have the slopes predicted by the edge-class rule, Eq. (9).

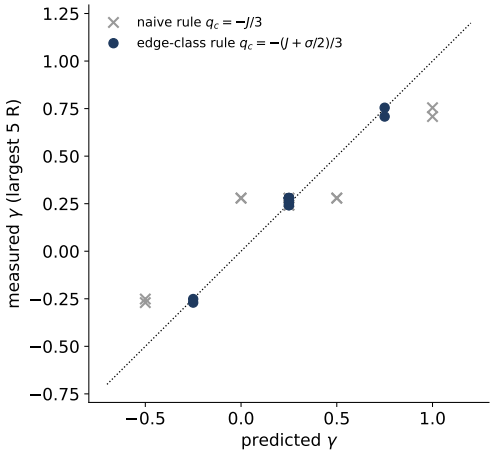


FIG. 13. Measured against predicted corner exponents for all thirteen corners (hexagons, triangles, parallelograms): edge-class rule (circles) and naive rule (crosses).

ages keep their orientation, whereas free boundaries reverse it [12, 16, 17].

V. SUMMARY AND OUTLOOK

Table III summarizes what survives. The corner charge remains the central object. Its value is fixed by the boundary height, but on the honeycomb lattice the relevant height is the continuum boundary value, which differs from the lattice boundary height by a class-dependent shift. Image rules follow the boundary type, as established by Ciucu for lozenge tilings. Breaking bipartiteness removes the Coulomb interaction beyond a length of order $1/t$ while leaving the lattice-scale boundary structure.

Two questions remain open. The first is a closed-form evaluation of the shift $2\delta = 1/2$: the exact strip reduces it to a one-dimensional Wiener–Hopf problem for each wave number, which we have solved only numerically. The second is the boundary chemical potential of Paper I on weighted Aztec diamonds, whose arctic curve is an ellipse. A further consequence of Eq. (2) is testable with the same Monte Carlo: a fixed charge binds a mobile defect when the local exponent falls below -2, which for a single vacancy happens at $g > 2$ in the bulk but already at $g > 4/3$ next to a straight edge.

Appendix A: Numerical methods

Pfaffian orientation. For the lattice of Sec. III we label sites (r, c) with $y = r$ upward. A lower-right triangle $(r, c) \to (r, c+1) \to (r+1, c+1)$ has an odd number of clockwise edges if the diagonal points north-east exactly when the vertical edge of column $c + 1$ points up; the upper-left triangle requires the vertical edges of neighbouring columns to alternate. Square faces at $t = 0$ are then automatically Kasteleyn. Deleting an interior vertex creates a face that violates the condition; reversing the signs of the edges crossed by the half-line $x = c + \epsilon$, $y < r$ repairs it and leaves all other faces unchanged. For pairs we use either two downward strings or a horizontal string between the monomers.

Honeycomb. With all weights $+1$ every hexagonal face satisfies the Kasteleyn condition. A deleted interior vertex merges three hexagons into a twelve-sided face, which is repaired by a downward string. Edge probabilities are $P_{ab} = K_{ab} K^{-1}_{ba}$.

Precision. For pair correlations, rank-one Schur-complement updates of a single inverse lose digits when $G(u, v) \sim e^{-\kappa r}$ is small. We therefore compute $\log|\det|$ of the correct string-modified matrix directly by sparse LU (SuperLU with COLAMD ordering) for every separation; this agrees with dense determinants to 10^{-11} for all t. Mean heights are integrated on the dual graph with curl errors below 4×10^{-15}. Codes and data are in the `paper2` branch of the repository accompanying Paper I.

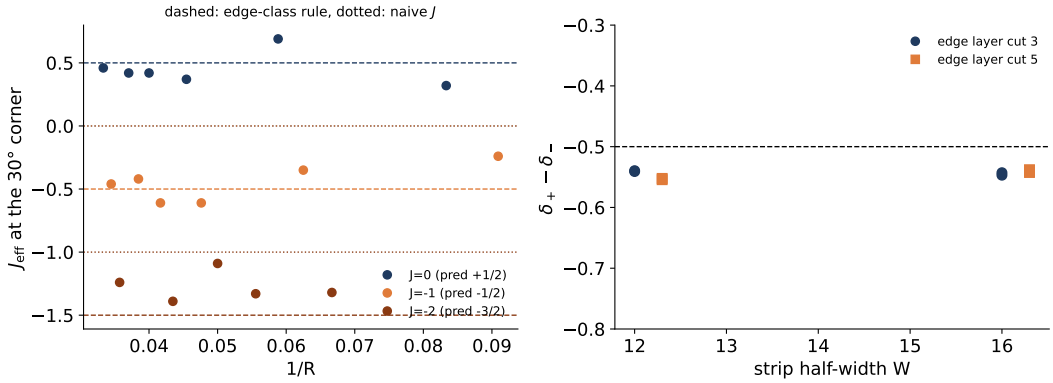


FIG. 14. (a) Effective jump at the 30° corner of balanced hexagons, from fits of the exact interior mean height; dashed lines are the edge-class rule, dotted lines the naive integers. (b) Edge-class shift measured on long strips.

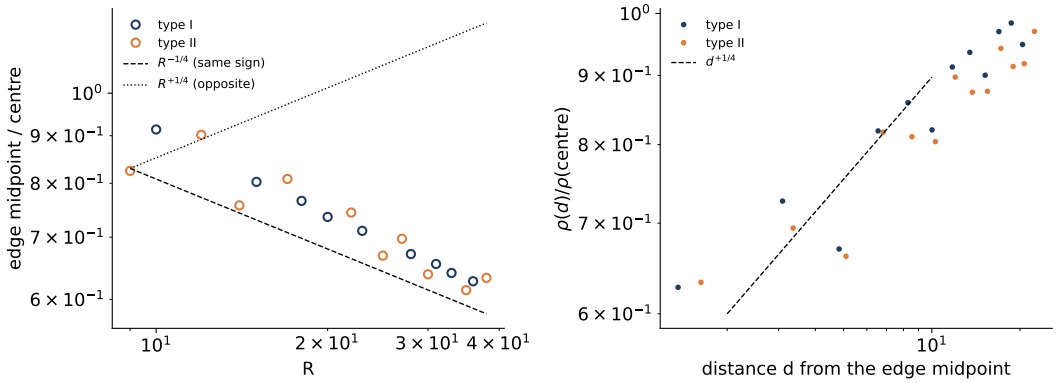


FIG. 15. Same-sign images on armchair edges. (a) Edge-midpoint to centre ratio against size; dashed: $R^{-1/4}$ (same sign), dotted: $R^{+1/4}$ (opposite sign). (b) Inward profile from the edge midpoint.

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TABLE III. Verdicts on the universality hypotheses (this work and Paper I).

quantity	hypothesis	verdict
corner exponent γ	lattice-independent at $g = 1/2$; scales with g	confirmed at $g = 1/2$ on square and honeycomb lattices (both corner angles); g dependence derived, Eq. (3), and confirmed by Monte Carlo (edge/bulk ratio 2.01 ± 0.05 for $0.09 \leq g \leq 1.9$)
image sign	set by the boundary type	same sign on armchair edges; constrained/free rule known [12, 16, 17]
corner charge q_c	boundary height jump / winding	honeycomb: $w = 3$; exact at 60°; 120° corners need the edge-class shift, Eq. (9) (blind test passed)
boundary layer	non-universal	confirmed; on the honeycomb it shifts the continuum boundary height by $\pm 1/4$ (sign by symmetry, size to 5×10^{-4})
non-bipartite defects	deconfined, screened	bulk known exactly [19, 20] and reproduced; boundary trapping stops at $N \sim 1/t$ with angle-dependent saturation; closed form at the critical point $t = 1/2$, derived from the rational symbol
density scaling function	conformal map (flat)	not yet compared in full on the honeycomb
μ, layer cost	free-fermion models	not tested here